# Benjamin David Smith

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## Research Interests

Warm-atomic vapors, nonlinear optics, laser cooling and trapping, quantum optics, quantum sensing, atom interferometry, quantum memories & communication, optical magnetometry, Bose-Einstein condensates, sonoluminescence, physics education research.

## Education

2016 – 2024 **Doctor of Philosophy in Physics**
University of Alberta, Department of Physics
Edmonton, AB, Canada

- Supervisor: Professor Lindsay J. LeBlanc
- GPA: 3.64/4.00.
- Dissertation: *Nonlinear interactions using neutral atomic gases* [link]

2009 – 2015 **Bachelor of Science in Physics, *cum laude***
Brigham Young University, Department of Physics and Astronomy
Provo, UT, USA

- Minors: Mathematics, Music
- GPA: 3.87/4.00.
- Senior Thesis: *Reflectance Measurements of Two Thin Film Materials in the UV* [link]

## Research Positions

2026 – Present **NRC Research Fellow**
Air Force Research Laboratories

- Experimental atom interferometry with Bose-Einstein condensates in fiber Fabry-Perot microcavities.

**Mentor**: Dr. Spencer Olson

2024 – 2026 **Postdoctoral Research Associate**
Sandia National Laboratories

- Experimental quantum-enhanced gravity and acceleration sensing using cold atom matter-wave interferometry, with applications toward practical inertial guidance and navigation.

**Supervisor**: Dr. Jongmin Lee

2016 – 2024 **Graduate Research Assistant**
University of Alberta, Department of Physics

- Upgraded a warm-atom microwave cavity system for experiments in nonlinear optics, microwave-to-optical conversion, and magnetometry.
- Developed 'Spinor-GPE', an open-source Python package for GPU accelerated solutions to the two-component, nonlinear Schrödinger equation, with applications for spin-orbit coupled Bose-Einstein condensates (BECs). [link]
- Operated an ultracold $^{87}$Rb BEC apparatus for experiments in quantum memory and spin-orbit coupling.

**Supervisor**: Professor Lindsay J. LeBlanc [link]

2014 – 2015 **Undergraduate Research Assistant**
Brigham Young University, Department of Physics and Astronomy

- Refurbished a discharge light source for generating the 30.4 nm extreme ultraviolet (EUV) helium emission line.
- Recommissioned a monochrometer for vacuum ultraviolet reflectance measurements of thin film aluminum
- Collected EUV reflectance data of thin film $Y_2O_3$ samples at the Advanced Light Source; fit a reflectance model to extract the film optical constants.

**Supervisors**: Professors R. Steven Turley & David D. Allred

2013 – 2014 **Undergraduate Research Assistant**

Brigham Young University, Department of Physics and Astronomy

- Calculated and tabulated thermal position distributions of crystal ions in various point symmetries.

**Supervisor**: Professor Branton J. Campbell [link]

## Teaching Experience

2016 **Adjunct Principle Instructor** (Brigham Young University - Idaho)

PHYS 150: Beginning Physics Lab (Winter & Spring 2016).

- Avg. student rating: 6.1/7.0.

PHYS 121: Principles of Physics 1 (Spring 2016).

- Avg. student rating: 6.3/7.0.

PHYS 220: Principles of Physics 3 (Winter 2016).

- Avg. student rating: 5.6/7.0.

2016 - 2023 **Teaching Assistant** (University of Alberta)

PHYS 295: Experimental Physics I (2023 & 2017).

PHYS 292: Experimental Physics for Engineers (2018).

PHYS 372: Quantum Mechanics A (2017).

PHYS 130: Wave Motion, Optics, and Sound (2016).

2014 - 2015 **Teaching Assistant** (Brigham Young University)

PHYS 245: Experiments in Contemporary Physics (2015).

PHYS 108: General Physics Lab 2 (2014).

PHYS 123: Introduction to Waves, Optics, and Thermodynamics (2014).

## Peer-Reviewed Publications

5 Babaei, B., **Smith, B.D.**, Tretiakov, A., Narayanan, A., & LeBlanc, L.J. "Microwave-optical double-resonance vector magnetometry with warm Rb atoms," *Applied Physics Letters* **128**, 184003 (2026).

4 **Smith B.D.**, Babaei, B., Narayanan, A., LeBlanc, L.J., "Microwave-to-optical conversion in a room-temperature $^{87}$Rb vapor with frequency-division multiplexing control," *Communications Physics* **6** 338 (2023).

3 **Smith, B. D.**, Cooke, L. W., & LeBlanc, L. J. "GPU-accelerated solutions of the nonlinear Schrödinger equation for simulating 2D spinor BECs," *Computer Physics Communications* 275 (2022): 108314.

2 Saglamyurek, E., Hrushevskyi, T., Rastogi, A., Cooke, L.W., **Smith, B.D.** & LeBlanc, L.J., "Storing short single-photon-level optical pulses in Bose–Einstein condensates for high-performance quantum memory," *New Journal of Physics* 23.4 (2021): 043028.

1 Muhlestein, J.B., **Smith, B.D.**, Miles, M., Thomas, S.M., Willey, A., Allred, D.D. & Turley, R.S., "$Y_2O_3$ optical constants between 5 nm and 50 nm," *Optics Express* 27.3 (2019): 3324-3336.

## Submitted Publications

1 **Smith, B.D.**, Orozco, A., & Lee, J. "A Compact, High-Data-Rate, High-Sensitivity GMOT-Based Light-Pulse Atom Interferometer toward Time-Multiplexed, Multi-Axis Inertial Sensing," *arXiv preprint* arXiv:2610.01735v1.

## Professional Memberships

2018 – Present **American Physical Society (APS)**

- **Divisions**:
  - Atomic, Molecular & Optical Physics (DAMOP)
  - Quantum Information (DQI)
- **Forums**:
  - Early Career Scientists (FECS)
  - Education (FEd)
  - History and Philosophy of Physics (FHPP)
  - Industrial and Applied Physics (FIAP)

## Awards and Scholarships

2022 **Oral Presentations, First Prize**
*13th Annual Symposium for Graduate Physics Research; Graduate Physics Student Association, University of Alberta.*
- Value: $175 CAD.

2018 **The Queen Elizabeth II Graduate Scholarship - Doctoral level**
*The Government of Alberta*
- Value: $7,500 CAD.

2016 **Doctoral Recruitment Scholarship**
*University of Alberta Faculty of Graduate Studies and Research*
- Value: $20,000 CAD.

2016 **Dean's Excellence Recruitment Scholarship Award**
*University of Alberta Faculty of Science*
- Value: $5,000 CAD.

2014 **Alvina Soffel Barrett Undergraduate Academic Scholarship**
*Brigham Young University*
- Value: $2,500 USD.

2009 **Brigham Young Undergraduate Academic Scholarship**
*Brigham Young University*
- Value: $4,290 USD.

## Grants Awarded

2025 **American Physical Society, Division of Quantum Information**
Wiki Scientist Course: Quantum Science
- Value: $500 USD.

2019 **NVIDIA Data Science GPU Grant**
Applied joint with fellow graduate student, Logan Cooke.
- Value: $3,000 CAD, in-kind.

2019 **APS DAMOP Conference Travel Grant**
- Value: $500 USD.

2015 **BYU Office of Research and Creative Activities Grant**
- Value: $1,500 USD.

## Proposals Submitted

2025 **National Academies of Sciences, Engineering, and Medicine: NRC Research and Fellowship Programs**

- Title: "*Realizing and Probing Strongly Interacting 1D Quantum Gases on a Hybrid Atom Chip–Cavity Platform*"
- Status: Awarded.

2025 **Sandia National Laboratories: Laboratory Directed Research and Development Exploratory Express**

- Title: "*Rapid, low-SWaP production of ultracold atoms for enhanced quantum sensing.*"
- Status: Not Awarded.

## Invited Oral Presentations

May 2025 "Intro to Quantum Communication: Principles, Applications, and Challenges."
*Sandia National Laboratories, RF Communications Department.* (Albuquerque, NM).

Feb. 2024 "Broad-tunable microwave-to-optical conversion in a room-temperature atomic vapor."
*Brigham Young University - Idaho, Physics Department Colloquium.* (Rexburg, ID).

Feb. 2024 "Broad-tunable microwave-to-optical conversion in a room-temperature atomic vapor."
*University of New Mexico, CQuIC Seminar.* (Albuquerque, NM).

Feb. 2024 "Broad-tunable microwave-to-optical conversion in a room-temperature atomic vapor."
*Los Alamos National Lab, MPA-Q.* (Los Alamos, NM).

Feb. 2024 "Broad-tunable microwave-to-optical conversion in a room-temperature atomic vapor."
*Sandia National Lab, AO Sensing Department.* (Albuquerque, NM).

Jun. 2023 "Microwave-to-optical conversion in a room-temperature atomic vapor."
*Sandia National Lab, AO Sensing Department.* (Albuquerque, NM).

## Contributed Conference Oral Presentations

Jun. 2026 — "A compact, high data rate, and high sensitivity cold atom interferometer."
*57th Annual Meeting of the APS DAMOP.* (Providence, RI).

Aug. 2023 — "Microwave-to-optical conversion in a room-temperature atomic vapor."
*Quanta CREATE Student Symposium.* (Edmonton, AB).

Jun. 2023 — "Broadly tunable and coherent microwave-to-optical conversion in a thermal atomic vapor."
*54th Annual Meeting of the APS DAMOP.* (Spokane, WA).

Oct. 2022 — "Broad-tunable, multimode nonlinear generation in a warm 87-Rb atomic vapor." [link]
*13th Annual Symposium for Graduate Physics Research; Graduate Physics Student Association, University of Alberta.* (Edmonton, AB).

Jun. 2022 — "Quantum Mechanics and You': Connecting to the Everyday." [link]
*NanoCanada, From Earth to Space.* (Edmonton, AB).

Jun. 2019 — "Synthetic electromagnetic forces in ultracold atoms." [link]
*50th Annual Meeting of the APS DAMOP.* (Milwaukee, WI).

Jul. 2018 — "How to engineer a BEC's kinetic energy: Techniques in quantum simulation."
*Quanta CREATE Student Symposium.* (Calgary, AB).

Oct. 2016 — "Making physics possible: The case for systematic assessment of learning and active learning practices in the university physics classroom."
*7th Annual Symposium for Graduate Physics Research; Graduate Physics Student Association, University of Alberta.* (Edmonton, AB).

Mar. 2015 — "Plasma light source for extreme ultraviolet reflectance." [link]
*BYU College of Physical and Mathematical Sciences Student Research Conference.* (Provo, UT).

Mar. 2014 — "Representational analysis of thermal vibrations in crystals." [link]
*BYU College of Physical and Mathematical Sciences Student Research Conference.* (Provo, UT).

## Other Oral Presentations

Aug. 2023 "Microwave-to-optical conversion in a room-temperature atomic vapor."
*U. of Alberta, GPSA Student Summer Seminar Series.* (Edmonton, AB).

Jul. 2020 "Intro to Driven Quantum Systems (AKA 'Shaking things up')."
*Quanta CREATE Tutorial Seminar.* (Virtual).

Aug. 2019 "Gauges, GPEs, and GPUs."
*U. of Alberta, GPSA Student Summer Seminar Series.* (Edmonton, AB).

Jun. 2017 "A Quantum Particle's Search for Meaning."
*U. of Alberta, GPSA Student Summer Seminar Series.* (Edmonton, AB).

## Contributed Conference Poster Presentations

Jun. 2025 "Cold atom systems for deployable quantum inertial navigation and gravitational sensing."
*56th Annual Meeting of the APS DAMOP.* (Portland, OR).

Jan. 2023 "The Doppler effect in a warm atomic vapor extends microwave to optical conversion tunability over a 500 MHz range of optical frequencies."
*Deep Tech Canada Quantum Days.* (virtual).

Jul. 2022 "Multimode microwave-to-optical conversion in warm $^{87}$Rb atoms." [link]
*The 27th International Conference on Atomic Physics.* (Toronto ON).

Jun. 2021 "Toward a spin tensor momentum coupled Bose Einstein condensate." [link]
*Canadian Association of Physicists (CAP) Congress.* (virtual).

Jun. 2021 "Toward a spin tensor momentum coupled Bose Einstein condensate." [link]
*52nd Annual Meeting of the APS DAMOP.* (virtual).

Jun. 2020 "Gauge fields and superfluid dynamics of ultracold atoms." [link]
*51st Annual Meeting of the APS DAMOP.* (virtual).

Aug. 2019 "AC synthetic gauge potentials in ultracold atoms."
*International Conference on Quantum Fluids and Solids.* (Edmonton, AB).

Jun. 2019 "AC synthetic gauge potentials in ultracold atoms." [link]
*50th Annual Meeting of the APS DAMOP.* (Milwaukee WI).

Jul. 2018 "Quantum Simulation via a Raman-coupled BEC of 87-Rb."
*Quanta CREATE Student Symposium.* (Calgary AB).

Jun. 2018 "Spin-orbit coupling and superfluidity in ultracold quantum gases." [link]
*49th Annual Meeting of the APS DAMOP.* (Ft. Lauderdale FL).

## Certifications & Professional Development

2025 **Wiki Scientist Course 2025: Quantum Science** [link]

2025 **Communications 303: Technical Proposal Writing**

- Course subjects: Discovering and reading calls; developing strategic proposal plans; drafting skills [8 hours].

Sandia National Laboratories; Albuquerque, NM

2022 **Graduate Teaching and Learning Program** [link]:

- Level 1 (*Foundations*) [25 hours].
- Level 2 (*Practicum*) [18 hours].
- Level 3 (*Pedegogy and Course Design*) [39 hours].

Faculty of Graduate Studies and Research, University of Alberta; Edmonton, AB

## Volunteer & Community Engagement

2026 **Session Chair** Matter-Wave Optics and Interferometry.
*57th Annual Meeting of the APS DAMOP.* (Providence, RI).

2022 **Workshop Instructor**: Quanta CREATE Student Symposium, "Quantum Computing Challenges & Exploration" [link]

- Developed and led an accessible and interactive instructional workshop for exploring quantum computing topics using IBM's open-source Qiskit software package and free interactive textbook.

2022 **Member, Organizing Committee**: Quanta CREATE Student Symposium

- Responsible for budgeting and securing external grant funding from FGSR (U. of Alberta) [$1000] and Alberta Innovates [$5000].
- Developed judging rubrics for student posters and talks.
- Created and maintained event website [Link].

2019 & 2022 **Volunteer, Presenter**: USchool

- Lead lab tours and optics demonstrations to groups of high school students.

2019 **Volunteer**: Quantum Fluids and Solids

- Prepared welcome materials for participants
- Microphone runner.

2015 **Volunteer, Camp Merit Badge Counsellor**: Camp Keowa, Boy Scouts of America

- Led boy scout participants in hands-on, interactive instruction in music and space exploration.

2015 **Volunteer, Judge**: Central Utah Science & Engineering Fair

- Adjudicated middle school science fair entries at a regional STEM competition.

2011 – 2013 **Volunteer, Full-time Missionary**: The Church of Jesus Christ of Latter-day Saints.

- Nevada Las Vegas Mission.

## Industry Experience

2015 **Product Specialist**: Qualtrics, LLC. – Provo, UT

- Instructed 20-30 remote academic users per day in professional online survey software use.
- Communicated complex instruction sets to resolve users' issues and meet their goals.

## Technical Skills

**Programming languages**

- Proficient in: Python, Markdown.
- Familiar with: HTML, Mathematica, C++, serial protocol.

**Hardware**

- Proficient in: Optics systems, laser stabilization, closed-loop controllers, signal acquisition & processing.
- Familiar with: Vacuum technologies, RF & microwave electronics.

**Software**

- Proficient in: LaTeX, Git, Autodesk Inventor, Inkscape.
- Familiar with: Zemax, LabVIEW, Sphinx, Hugo, Blender.

**Languages**

- English (fluent), Spanish (intermediate).

## Dedication

"*O give thanks unto the Lord; for he is good: because his mercy endureth for ever.*"

~ Psalms 118:1

"*And ye cannot bear all things now; nevertheless, be of good cheer,*
*for I will lead you along.*"

~ Doctrine & Covenants 78:18